Project Title

Exploratory Data Analysis of Data Science Job Market

Introduction

- Data Science is one of the fastest growing fields in the technology industry. Many companies hire professionals such as Data Scientists, Data Analysts, and Machine Learning Engineers. Salaries, job roles, experience levels, and work types can vary across companies and countries.
- In this project, we perform Exploratory Data Analysis (EDA) on a dataset related to data science jobs. The purpose of this project is to understand the job market, salary trends, and work patterns in the data science field using Python libraries such as NumPy, Pandas, Matplotlib, and Seaborn.

Problem Statement

- The data science job market is growing rapidly, but there is limited understanding about how factors like experience level, company size, job role, and remote work affect salaries and job opportunities.
- This project aims to analyze job data to find patterns and insights that help understand the data science job market and salary trends.

Objectives of the Project

The main objectives of this project are:

- To understand the structure of the dataset.
- To clean the data by handling missing values and duplicates.
- To analyze salary distribution in data science jobs.
- To study the relationship between experience and salary.
- To analyze job roles and company characteristics.
- To visualize important patterns using graphs.
- To generate insights and recommendations from the data.

Dataset Description

The dataset contains information about different data science jobs.

Some important columns in the dataset are:

- Job Title – Role of the employee (Data Scientist, ML Engineer, etc.)
- Experience Level – Entry, Mid, Senior, Executive
- Company Size – Small, Medium, Large
- Company Location – Country where the company is located
- Employment Type – Full-time, Contract, Part-time
- Remote Ratio – Percentage of remote work (0, 50, 100)
- Years of Experience – Total experience of employee
- Salary (USD) – Salary of employee in USD

The dataset contains both categorical and numerical data, which makes it suitable for EDA.

Tools and Libraries Used

The following Python libraries are used in this project:

- NumPy – For numerical operations
- Pandas – For data handling and analysis
- Matplotlib – For basic data visualization
- Seaborn – For advanced data visualization

Import Libraries

- First we import the required Python libraries.

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

- These libraries help us load, clean, analyze, and visualize the dataset.

Load the Dataset

In [2]:
```
df = pd.read_csv("data_science_jobs_eda_dataset.csv")
```

In [3]:
```
df.head()
```

Out[3]:

	job_title	experience_level	company_size	company_location	employment_type	re
0	AI Engineer	Mid	Medium	UK	Full-time	
1	Business Intelligence Analyst	Mid	Small	UK	Full-time	
2	Data Analyst	Senior	Small	USA	Full-time	
3	Business Intelligence Analyst	Senior	Medium	Canada	Full-time	
4	Business Intelligence Analyst	Mid	Large	Canada	Part-time	

Understand the Dataset

In [4]:
```
df.shape
```

Out[4]:
```
(2050, 8)
```

- Shows number of rows and columns.

In [5]:
```
df.columns
```

Out[5]:
```
Index(['job_title', 'experience_level', 'company_size', 'company_location',
       'employment_type', 'remote_ratio', 'years_experience', 'salary_usd'],
      dtype='object')
```

Dataset Information

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2050 entries, 0 to 2049
Data columns (total 8 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   job_title         2050 non-null   object 
 1   experience_level  2050 non-null   object 
 2   company_size      2050 non-null   object 
 3   company_location  1948 non-null   object 
 4   employment_type   2050 non-null   object 
 5   remote_ratio      2050 non-null   int64  
 6   years_experience  1949 non-null   float64
 7   salary_usd        1949 non-null   float64
dtypes: float64(2), int64(1), object(5)
memory usage: 128.3+ KB
```

- Shows data types and missing values.

Statistical Summary

In [7]:
```
df.describe()
```

Out[7]:

	remote_ratio	years_experience	salary_usd
count	2050.000000	1949.000000	1949.000000
mean	49.024390	7.051308	90429.780400
std	41.758846	4.287108	37781.701478
min	0.000000	0.000000	10395.000000
25%	0.000000	3.000000	62745.000000
50%	50.000000	7.000000	88269.000000
75%	100.000000	11.000000	114501.000000
max	100.000000	14.000000	207197.000000

- Provides mean, min, max, standard deviation etc.

Check Missing Values

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
job_title             0
experience_level      0
company_size          0
company_location    102
employment_type       0
remote_ratio          0
years_experience    101
salary_usd          101
dtype: int64
```

- This shows how many missing values exist in each column.

Handle Missing Values

- Fill missing values with median.

In [10]:
```
df["salary_usd"] = df["salary_usd"].fillna(df["salary_usd"].median())

df["years_experience"] = df["years_experience"].fillna(df["years_experience"].me
```

In [11]:
```
df["company_location"] = df["company_location"].fillna(df["company_location"].mc
```

- Missing values are replaced with median or most frequent value.

Check Duplicate Rows

In [12]:
```
df.duplicated().sum()
```

Out[12]:
```
np.int64(50)
```

In [13]:
```
df.drop_duplicates(inplace=True)
```

- Duplicate rows are removed to keep the dataset accurate.

Stored Cleaned Dataset

In [15]:
```
df.to_csv("Cleaned_Data.csv")
```

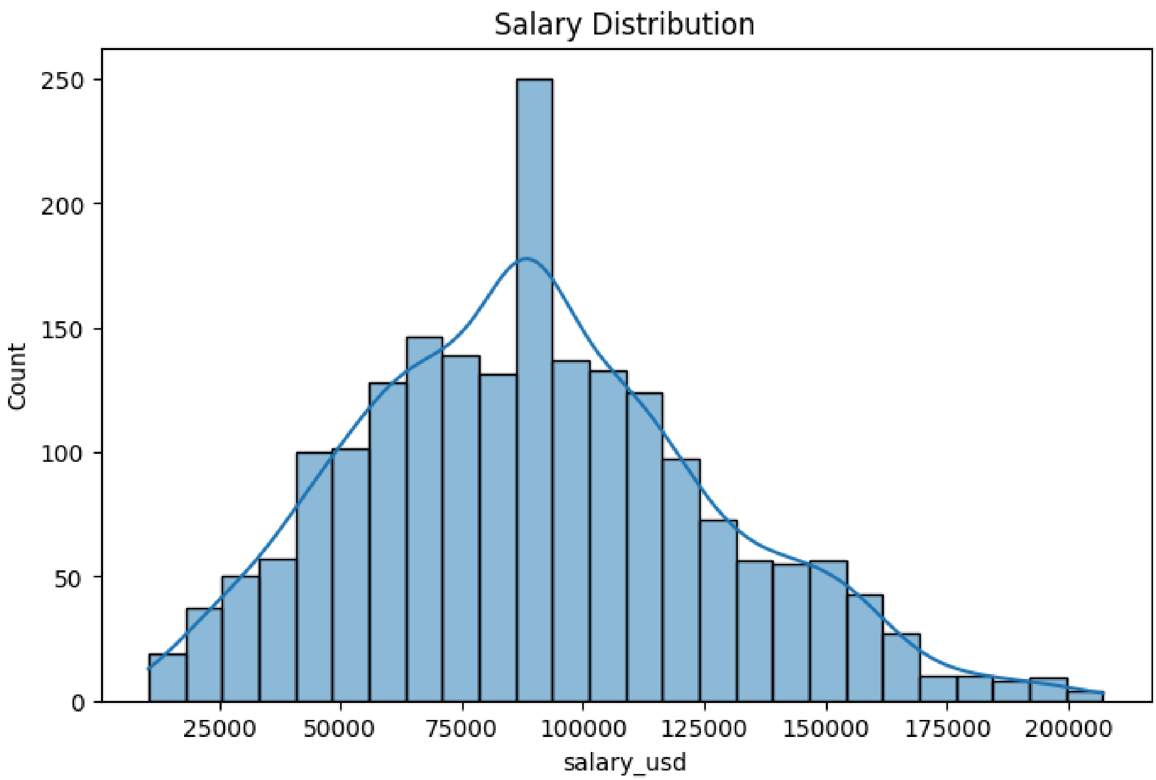
Import Cleaned Dataset For Analysis

In [16]:
```
df = pd.read_csv("Cleaned_Data.csv")
```

Analysis

Salary Distribution

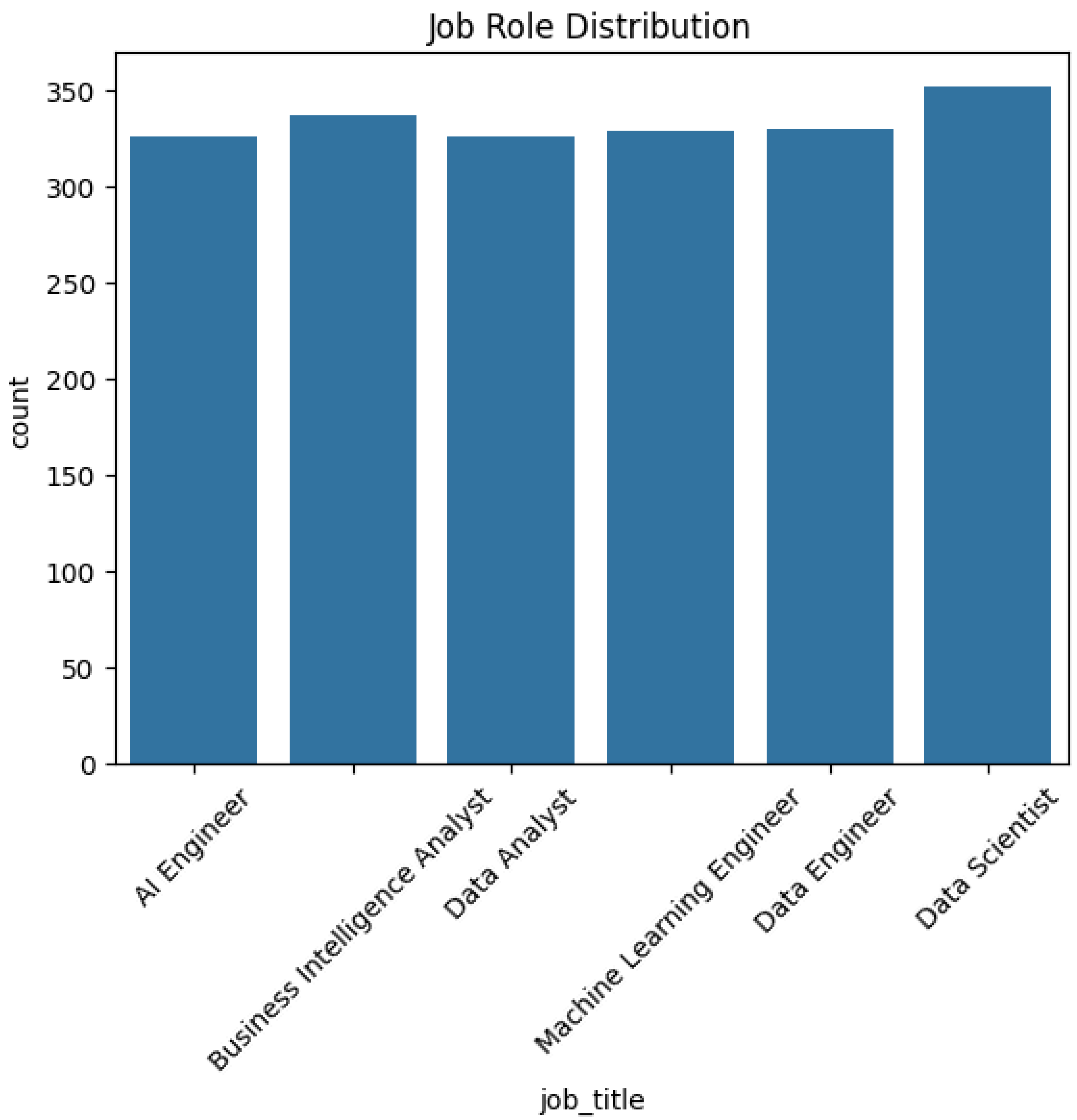
In [17]:
```
plt.figure(figsize=(8,5))
sns.histplot(df["salary_usd"], kde=True)
plt.title("Salary Distribution")
plt.show()
```



- Most salaries fall within a certain range and the distribution shows how salary values are spread.

Job Role Distribution

In [18]:
```
sns.countplot(x="job_title", data=df)
plt.xticks(rotation=45)
plt.title("Job Role Distribution")
plt.show()
```



- Shows which job roles appear most frequently in the dataset.

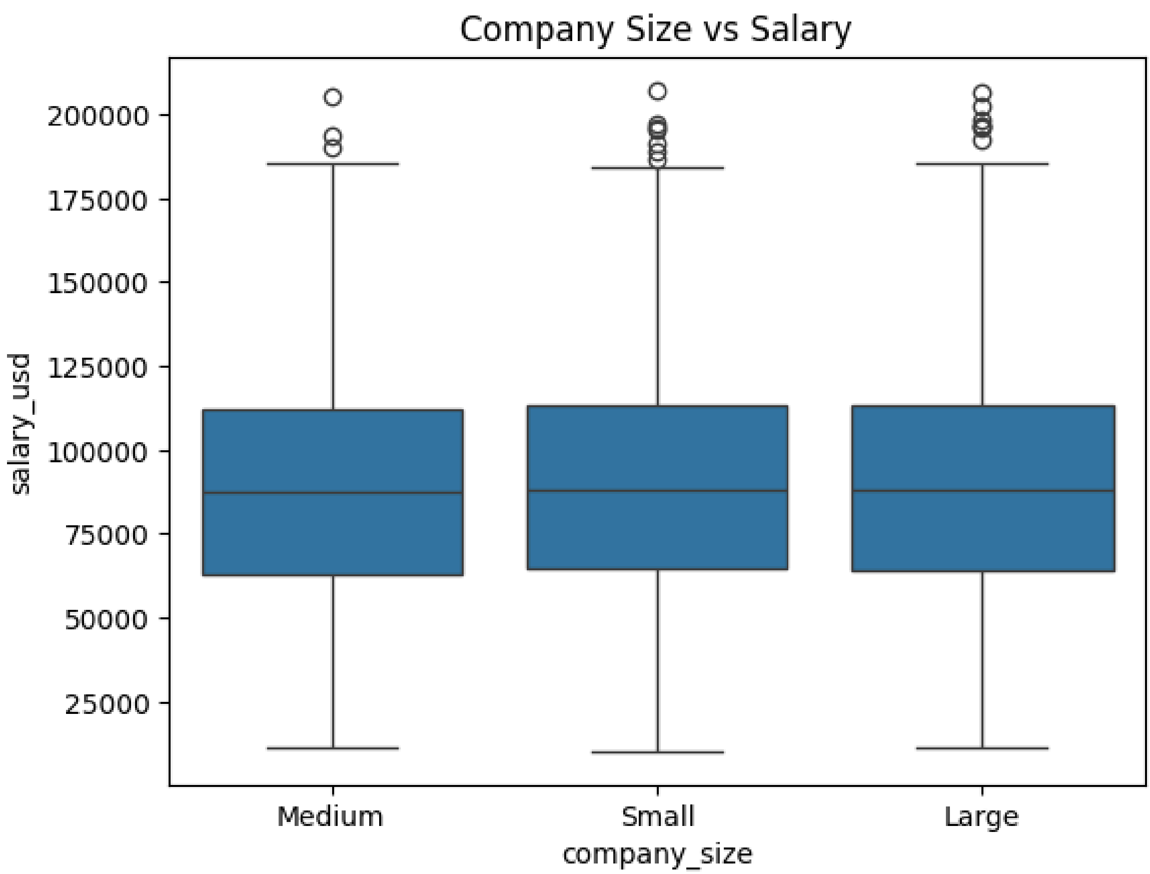
Experience vs Salary

In [19]:
```
sns.scatterplot(x="years_experience", y="salary_usd", data=df)
plt.title("Experience vs Salary")
plt.show()
```

- Salary tends to increase as years of experience increase.

Company Size vs Salary

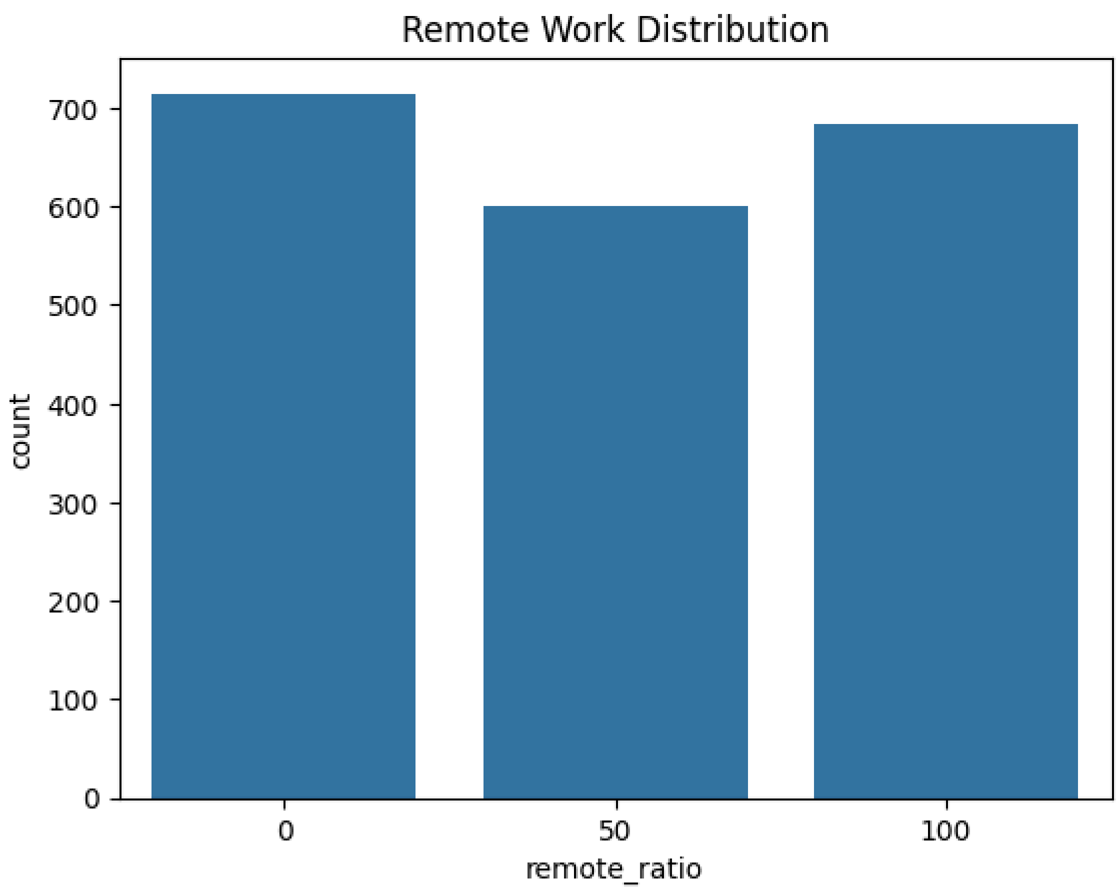
In [20]:
```
sns.boxplot(x="company_size", y="salary_usd", data=df)
plt.title("Company Size vs Salary")
plt.show()
```



- Large companies may provide higher salary packages.

Remote Work Analysis

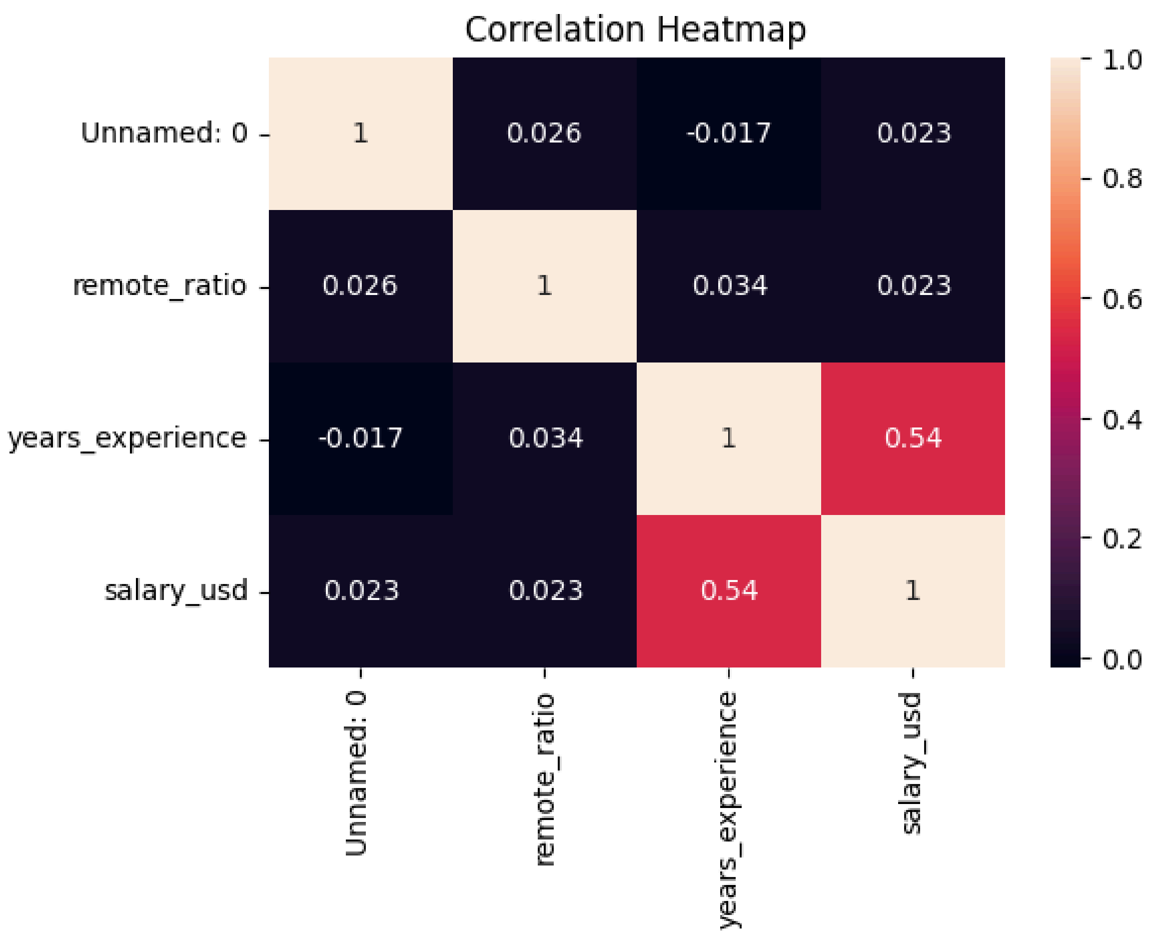
In [21]:
```
sns.countplot(x="remote_ratio", data=df)
plt.title("Remote Work Distribution")
plt.show()
```



- Shows how many jobs are onsite, hybrid, or fully remote.

Correlation Analysis

In [22]:
```
plt.figure(figsize=(6,4))
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



- Shows relationships between numerical variables like experience and salary.

Key Insights

- Salary increases with experience.
- Senior level roles earn higher salaries.
- Large companies tend to offer higher salaries.
- Remote work is common in data science jobs.
- Certain roles like Machine Learning Engineer have higher salaries.

Recommendations

- Data science professionals should focus on building experience and advanced skills.
- Companies should provide remote or hybrid work options to attract talent.
- Organizations should invest in employee training and development.

Conclusion

- This project performed Exploratory Data Analysis on a dataset related to data science jobs. The analysis helped understand salary trends, job roles, experience levels, and work patterns.

- Using Python libraries such as NumPy, Pandas, Matplotlib, and Seaborn, meaningful insights were discovered that help understand the current data science job market.